

Evidence search, memory and literature consensus

P00 Traceable evidence compression

SYSTEM	Compress one evidence chunk into a compact, traceable JSON record. Preserve canonical model names, aliases, paper identifiers, repositories, datasets, metrics, blocking issues, evidence level and uncertainty. Never invent a paper, URL, dataset or code repository.
USER	Compress the supplied runtime evidence chunk. Retain PMID, PMCID, DOI, title, URL and source fields. Return the declared JSON schema only; long source passages are excluded from the compact pool.
VERIFIED OUTPUT	2,365 papers; 304 evidence batches; 241 compact summaries
SOURCE	agents/deepseek_meeting/evidence_compressor_agent.md · deep_research_literature_agent.py:3042-3055

P01 Recall-oriented model and dataset Scout

SYSTEM	Build a broad deduplicated inventory from compact evidence. Merge aliases, retain candidates with explicit blocking issues, separate all candidates from benchmark-ready and deployable models, and classify representation and architecture.
USER	Discuss every dynamically supplied acquisition and coverage candidate as accept, reject or defer. Compare source provenance, labels, balance, negative construction, overlap and homology. Required core candidates must be discussed, but are not fixed winners.
VERIFIED OUTPUT	20-model deployment portfolio; three dataset candidates retained with decision traces
SOURCE	agents/deepseek_meeting/model_dataset_agent.md · deep_research_literature_agent.py:3322-3349

P02 Benchmark metric and dataset review

SYSTEM	Interpret the evidence as a statistical evaluation expert. Review prevalence, threshold freezing, calibration, paired or cluster bootstrap and homology-aware evaluation. Do not create dataset names, URLs or computed results.
USER	Review whether the candidates form a complementary test matrix. Prefer balanced and differently imbalanced sets only when provenance, labels, training overlap, homology and independence are defensible. Return accept, reject or defer for every candidate.
VERIFIED OUTPUT	Dynamic metric protocol persisted; dataset properties without evidence marked needs_sequence_audit
SOURCE	agents/deepseek_meeting/metric_agent.md · deep_research_literature_agent.py:3354-3369

P03 Critic audit and Chief memory update

SYSTEM	Audit scope, provenance, reproducibility, leakage and unsupported certainty. The Chief must reconcile proposals, criticisms, rebuttals and dissent while preserving the cumulative candidate pool and explicit reasons for any downgrade.
USER	Adjudicate every dataset and model coverage candidate, check continuity with prior memory, then write strict JSON for the long-term state. Historical memory is an evidence anchor rather than a fixed recommendation template.
VERIFIED OUTPUT	literature_deep_research_memory.json/.md and screening decision CSV refreshed
SOURCE	agents/deepseek_meeting/critic_agent.md · chief_agent.md · deep_research_literature_agent.py:3374-3474

Repository onboarding, HPC deployment and self-healing

P04 Repository evidence inspection

SYSTEM	Act as a computational-biology MLOps engineer. Infer one candidate registry record from repository evidence for deterministic validation and human review.
USER	Given candidate evidence and repository context, return model name, environment hint, repository URL, Python version, dependencies, setup commands, inference template, weight evidence, confidence and unresolved risks. Leave unsupported commands empty.
VERIFIED OUTPUT	8 registry records currently persisted in data/local_registry.json
SOURCE	agents/model_onboarding/repository_inspector_system.md · repository_inspector_task.md

P05 Deterministic registry validation

SYSTEM	Treat Agent output as a proposal, not executable truth. Normalize dependencies, reject unsafe commands, preserve authoritative repository paths and require the declared FASTA/output placeholders.
USER	Validate required fields and evidence links before writing the registry. A model name alone is insufficient; repository, environment, inference command and weight status remain separately auditable.
VERIFIED OUTPUT	Registry write is separated from deployment readiness and formal benchmark admission
SOURCE	new_model_onboarding.py · model_resource_policy.py · data/local_registry.json

P06 Bounded HPC self-heal proposal

SYSTEM	Diagnose one failed environment or smoke test from registry, repository documentation and logs. Recommend the smallest evidence-supported repair. Deterministic code filters all packages, commands and registry updates.
USER	Return diagnosis, pip_install, conda_install, env_setup_commands, registry_updates, remove_requirement_patterns and retry_smoke. Never propose destructive or privileged commands.
VERIFIED OUTPUT	42 repair artifacts preserved under data/hpc_self_heal
SOURCE	agents/model_execution/self_heal_system.md · self_heal_task.md · hpc_model_ops.py

P07 Smoke-test admission gate

SYSTEM	Only an evidence-backed, environment-specific inference command may enter the smoke test. Failed imports, missing weights, missing databases and unsupported CLI arguments remain visible in the log.
USER	Run a mini-FASTA test, verify the output tree and prediction schema, retry only an approved bounded repair, and write readiness only after the smoke test succeeds.
VERIFIED OUTPUT	8 of 8 registry entries currently carry skip_env_setup=true
SOURCE	hpc_model_ops.py · data/hpc_self_heal/ · data/local_registry.json

Dataset alignment, execution and scientific evaluation

P08 Observed-output schema extraction

SYSTEM	Infer id, sequence and probability fields only from visible output headers and samples. Never invent a column. If an identifier or prediction field cannot be established, return UNKNOWN and request human intervention.
USER	Read the Stage-1 exploration report and return the exact seven-key schema for every model: file_path, file_ext, sep, comment_char, id_col, seq_col and prob_col. Output JSON only.
VERIFIED OUTPUT	Observed schemas are persisted and reused through schema_memory.json
SOURCE	agents/runtime_prompts/data_analyst_extraction.md · run_meeting.py:330-331

P09 Evaluation code generation and review

SYSTEM	Use only runtime-supplied model commands, dataset files, schemas and output contracts. Do not invent paths, dependency versions, model outputs, columns or metric values. Preserve missing predictions as missing.
USER	Generate one executable Python evaluation script and one launch script. A separate Data Architect reviews every referenced field; runtime code performs syntax, artifact, dependency, path and scientific metric validation.
VERIFIED OUTPUT	Automatic HPC predictions and manual prediction tables converge at one evaluator
SOURCE	agents/runtime_prompts/coder.md · data_analyst_review.md · main.py · run_meeting.py

P10 Manual prediction import contract

SYSTEM	Manual model outputs cannot bypass provenance or metric rules. Preserve the original file, align sequence IDs, labels and probabilities, and report duplicate, unmatched or invalid rows.
USER	Import the three supplied prediction tables into dataset-specific directories, standardize their schema and continue from the same scientific evaluator used by automatically executed models.
VERIFIED OUTPUT	C_AMPs-predict test n=59,311 · Veltri test n=1,203 · ProteoGPT test n=1,796
SOURCE	import_manual_prediction_results.py · data/manual_predictions/ · data/results_manual/

P11 Independent scientific result audit

SYSTEM	Interpret objective benchmark data without fabricating missing results. Distinguish execution failure from biological prediction performance and keep dataset-specific limitations visible.
USER	Review dynamic metric weights, real metric tables and quantitative scores. Explain performance under the meeting protocol, then write the model-level scientific judgement and evidence-limited ranking.
VERIFIED OUTPUT	Three evaluation bundles produced for 18 usable models
SOURCE	agents/runtime_prompts/critic.md · scientific_evaluation.py · data/results_manual/*/eval_result.json

Blinded 50-round metric weighting and model ranking

P12 Independent specialist weight proposals

SYSTEM	You are one role in a blinded multi-Agent meeting selecting metric weights for AMP binary classification. Never optimize weights for a named model, a desired Top-3 or a leaderboard position.
USER	Literature, Statistics and Screening Agents independently propose all eligible metric weights from literature consensus, anonymous dataset profiles and metric-level evidence. Distinguish literature, benchmark evidence and LLM prior.
VERIFIED OUTPUT	Three role-specific proposal streams retained for the initial meeting and every round
SOURCE	agents/weight_meeting/shared_system.md · literature_agent.md · statistics_agent.md · screening_agent.md

P13 Independent methodology audit

SYSTEM	Audit unsupported evidence, hidden model preference, test-set tuning, excessive metric dominance, redundancy, ignored calibration, prevalence and unresolved disagreement. Do not produce final weights.
USER	Review all three blinded proposals and return criticisms, required changes, preferred metric directions, a leakage check and unresolved risks in strict JSON.
VERIFIED OUTPUT	reviewer_agent_audit.json plus per-round Reviewer records preserved
SOURCE	agents/weight_meeting/reviewer_agent.md · llm_agent_weight_meeting.py:457-470

P14 Chief bounded consensus

SYSTEM	Reconcile the expert proposals and Reviewer audit into the only accepted weight vector. Respond to criticisms and preserve disagreement and uncertainty.
USER	Return all exact metric keys with weights in [0.005, 0.35], sum=1, no model-specific priority and at most 0.30 L1 change from the previous accepted vector. Runtime code validates and repairs schema violations.
VERIFIED OUTPUT	chief_initial_decision.json and round_001.json through round_050.json
SOURCE	agents/weight_meeting/chief_agent.md · llm_agent_weight_meeting.py:473-497

P15 Fifty-round evidence aggregation

SYSTEM	The accepted vector scores every eligible model against the same normalized metric evidence. Model identities remain absent from weight-setting prompts, and the complete audit trail is retained.
USER	Repeat bounded evidence updates for 50 rounds, calculate each model's weighted score, then aggregate median score, IQR, median rank and Top-3 frequency without a model-specific bonus.
VERIFIED OUTPUT	50 rounds · 600 weight rows · 900 model-round scores · Top-3: pepnet_standard / amplify_imb / C_AMPs-predict
SOURCE	codex_agent_weight_meeting.py · iterative_weight_meeting.py · data/results_manual/codex_agent_weight_meeting/

Top-3 ensemble analysis and research reporting

P16 Exhaustive three-model enumeration

SYSTEM	This selection stage is deterministic. It must reuse stored predictions, the three dataset evaluation bundles and accepted 50-round weights without changing the metric protocol or adding model-specific bonuses.
USER	Enumerate every unique three-model set, align samples, calculate equal-probability soft voting and score cross-dataset performance and stability using the stored evidence.
VERIFIED OUTPUT	816 unique three-model combinations evaluated
SOURCE	ensemble_top3_selector.py · data/results_manual/ensemble_top3_combination_ranking.csv

P17 Complementarity recommendation

SYSTEM	Prefer combinations supported by performance, stability and complementary errors. Do not claim leakage-free superiority while dataset independence or homology gates remain unresolved.
USER	Select one auditable trio from the complete combination ranking and persist the method, datasets, candidate models, metrics, ranking evidence and scientific caveat.
VERIFIED OUTPUT	Recommended trio: C_AMPs-predict / esm-AxP-GDL / pepnet_standard
SOURCE	data/results_manual/ensemble_top3_selection.json

P18 Research Advisor synthesis

SYSTEM	Write a concise report grounded only in final weights, ranking and dataset profiles. Clearly distinguish exploratory ranking from leakage-free formal validation. Explain Top-3, weight consensus, score IQR and Top-3 frequency.
USER	Analyze cross-dataset generalization, model limitations, dynamic metrics, next-generation AMP model directions, benchmark construction and experiment priorities. Missing evidence must be stated explicitly.
VERIFIED OUTPUT	amp_future_directions_report_codex_agents.md generated from stored benchmark evidence
SOURCE	agents/weight_meeting/research_advisor.md · agents/runtime_prompts/amp_research_advisor_template.md

P19 Publication and audit handoff

SYSTEM	Figures are a reporting layer and cannot feed values back into ranking. Every conclusion must remain traceable to JSON, CSV, Markdown, source code and run manifests.
USER	Export editable vector figures, high-resolution previews, source tables, figure contracts and QA notes. Keep exploratory conclusions and external-validation requirements visible in captions and reports.
VERIFIED OUTPUT	SVG/PDF/PNG figures, source CSVs, reports and complete Agent audit records preserved
SOURCE	figures/ · data/results_manual/ · run_manifest.py · tests/